

Seeing space and time together: glyph maps in R

Line, interval and interactive glyph maps in ggplot2 with **cubble** and **sugarglider**

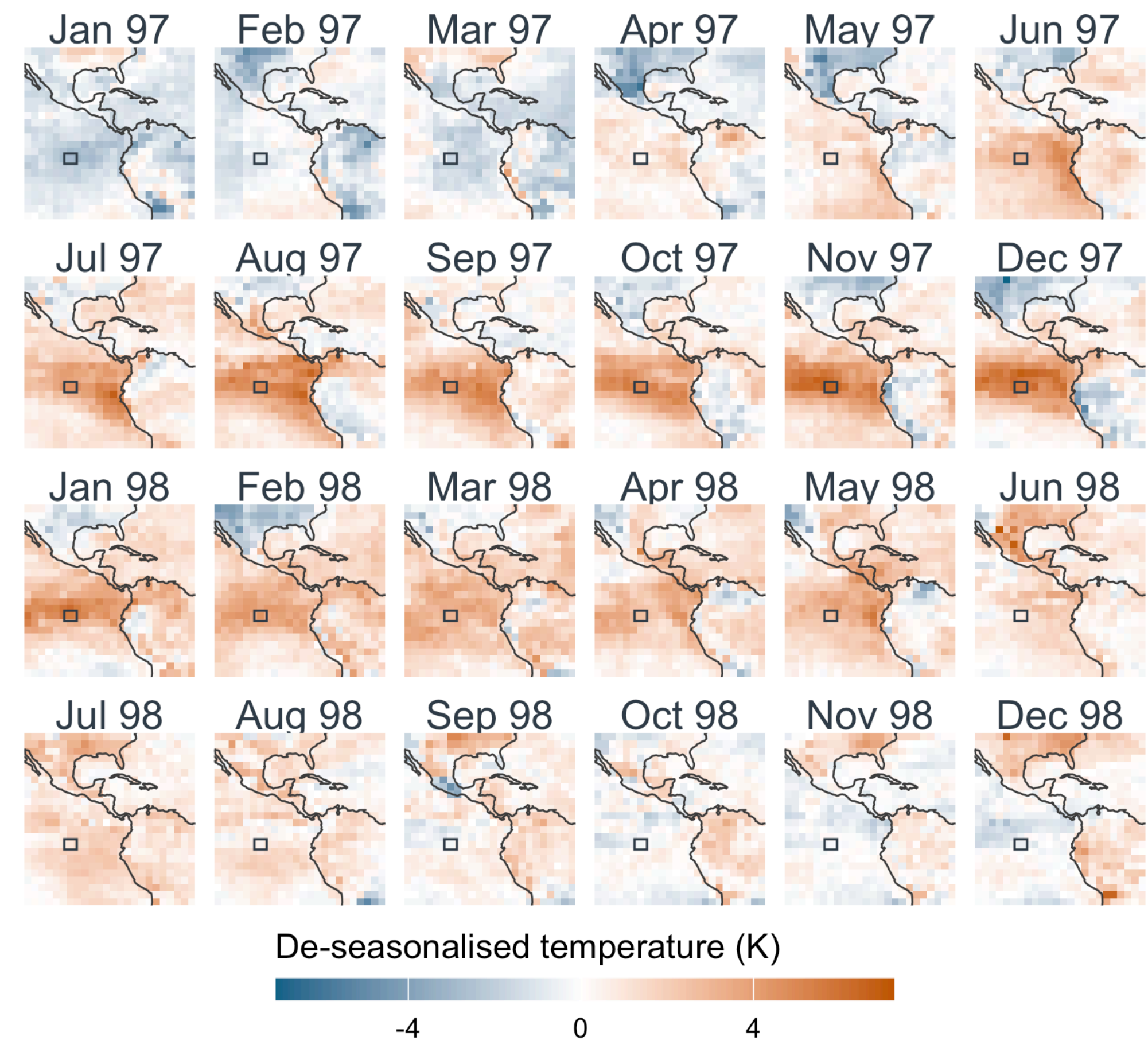
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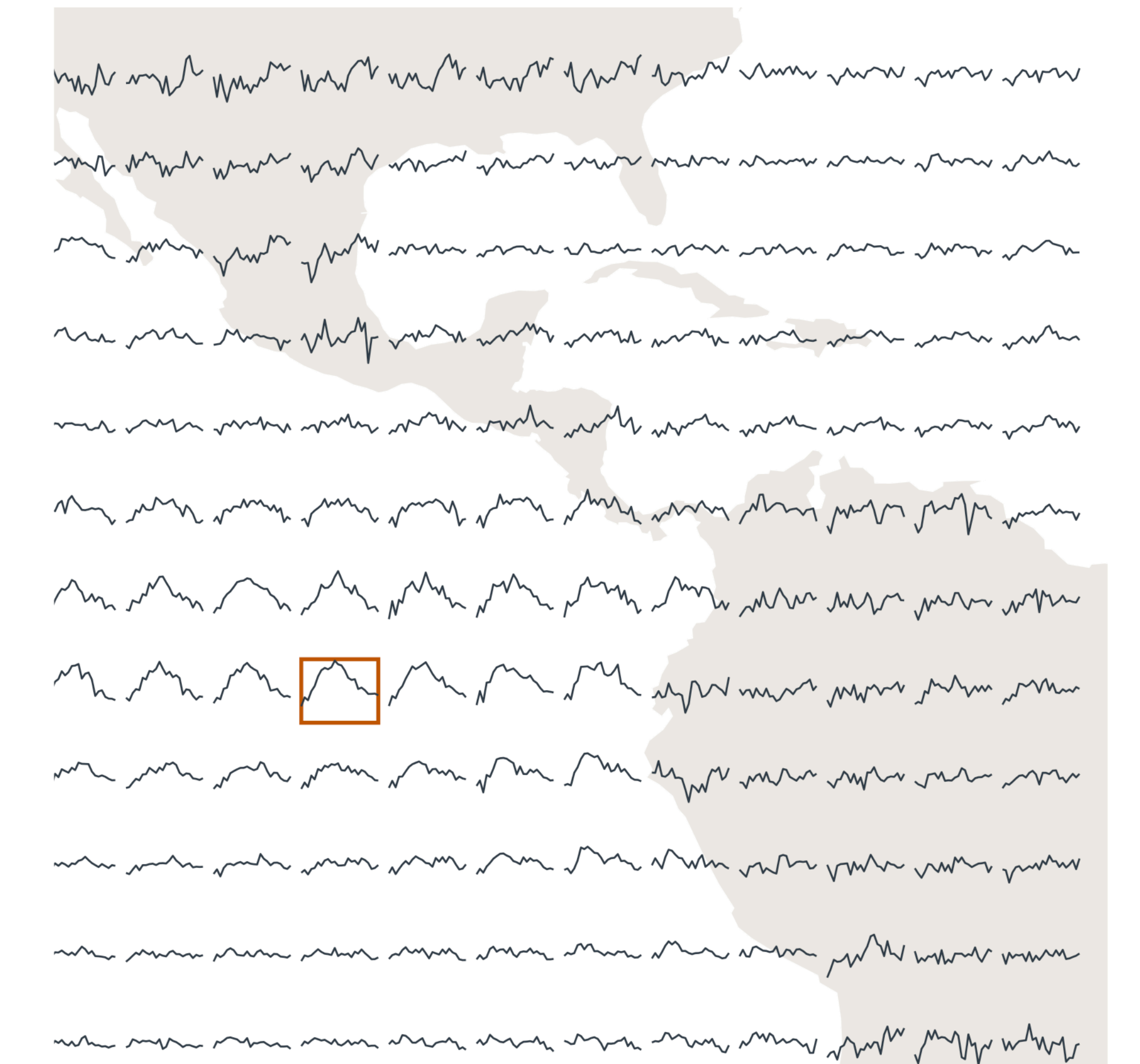
2012 From facets to a glyph map

Environmental data often come as **many time series at many locations**, commonly shown as maps faceted by time. Below, monthly de-seasonalised surface temperature over Central America for 1997–98.



Trace one location. Follow the boxed cell through the 24 panels to get its time series. A **glyph map** draws that series as a small plot at the cell's location (Wickham et al., 2012).

Do this for every location to get the full glyph map (`GGally::glyphs()` computes the glyph coordinates):

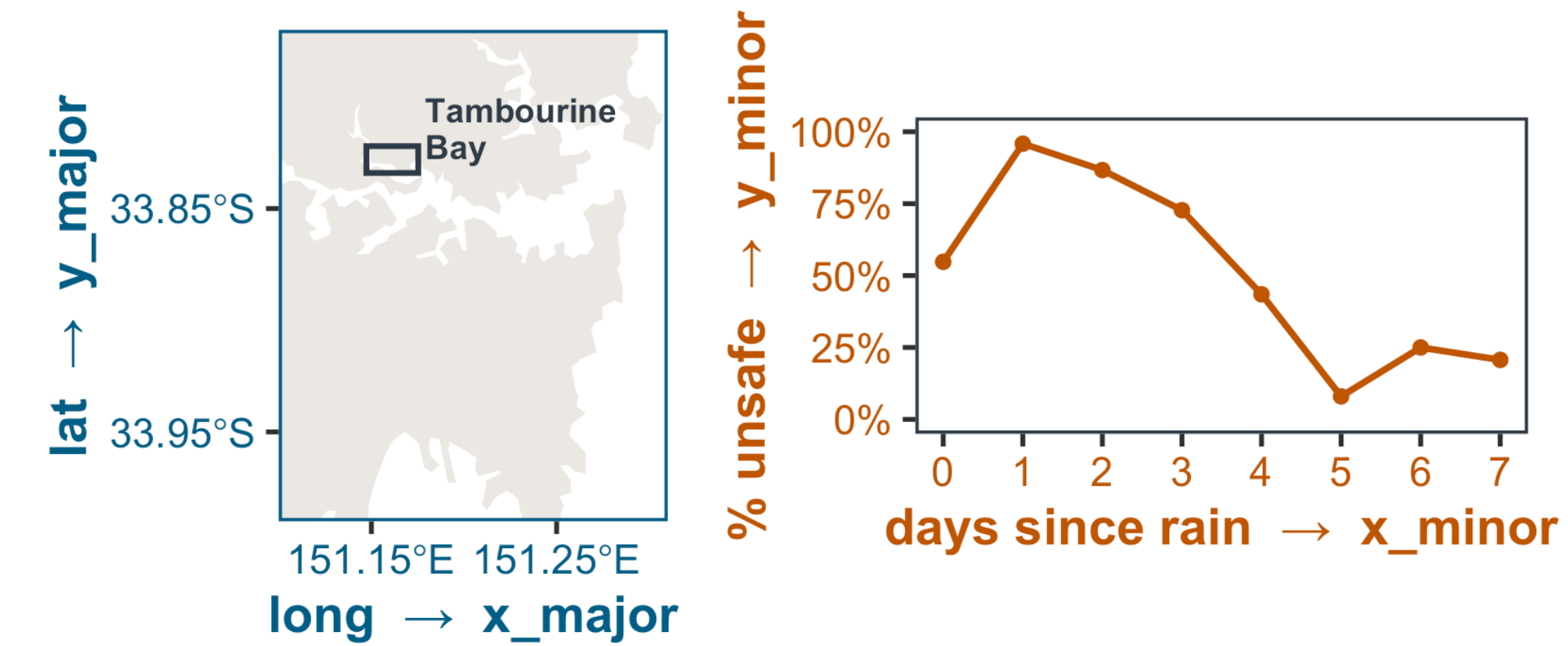


What the glyph map shows:

- Equatorial eastern Pacific glyphs share one hump peaking in late 1997 (the El Niño), fading away from the equator.
- Glyphs over land vary far more month to month than those over the sea.

2024 cubble: glyph maps in ggplot2

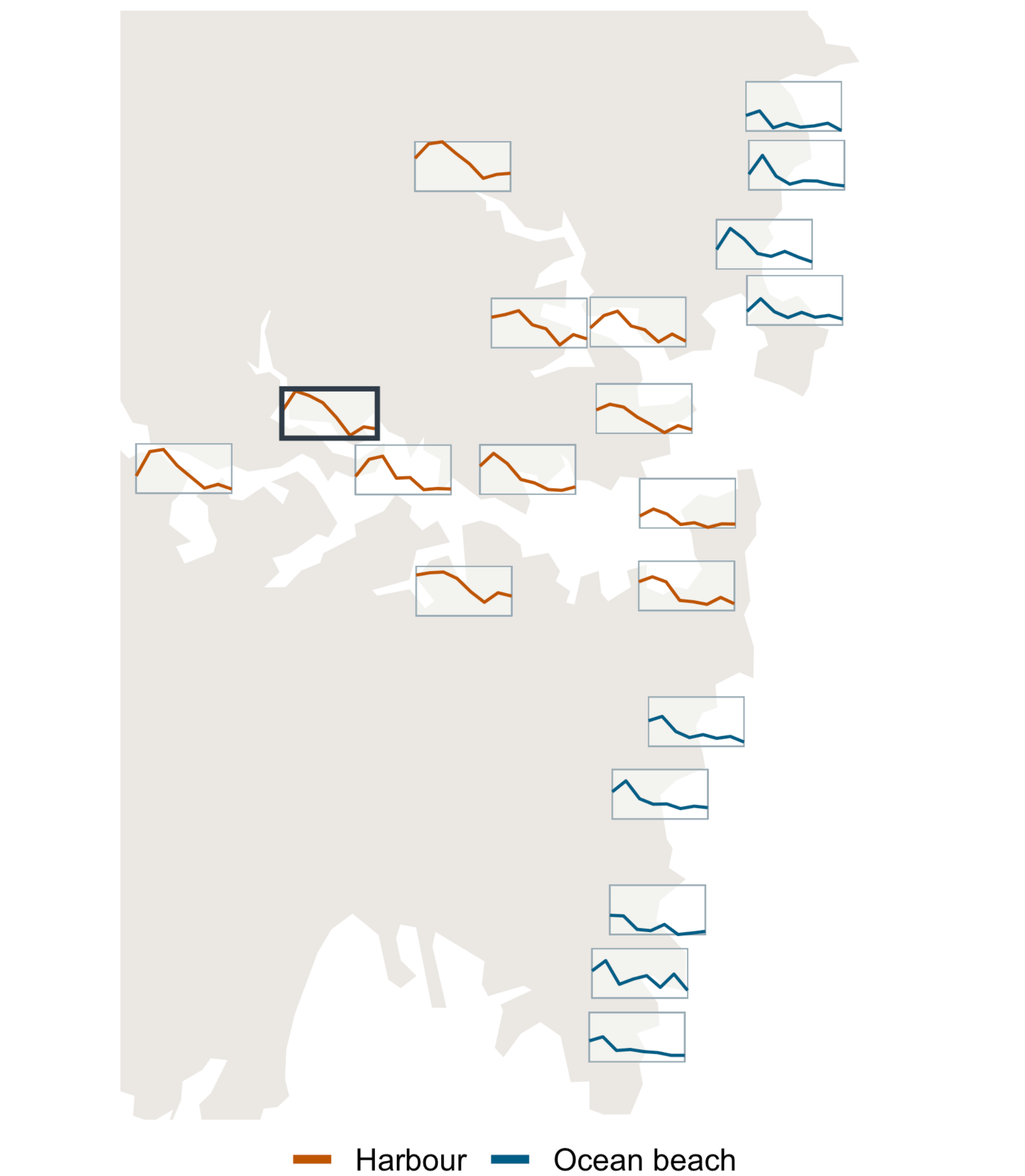
cubble implements the glyph-map transformation in ggplot2 syntax: `geom_glyph()` takes the major and minor coordinates as four new **aesthetics**, so making a glyph map is a matter of mapping your variables onto them. Here: **how long after a storm is it safe to swim in Sydney?** Share of water samples above the bacteria guideline (40 enterococci/100 mL) in the 7 days after ≥ 20 mm of rain, 1993–2025 (TidyTuesday 2025-05-20).



Tambourine Bay's location gives the **major** coordinates; its recovery curve after rain gives the **minor** coordinates drawn inside its box.

Because the glyph is an ordinary ggplot2 layer, it pieces together with other layers. The key parts of the code (simplified):

```
ggplot(beach,
  aes(x_major = long, y_major = lat,
      x_minor = days_after_rain,
      y_minor = unsafe,
      colour = water)) +
  geom_sf(data = sydney, inherit.aes = FALSE) + # map
  geom_glyph_box(width = 0.028, height = 0.012) +
  geom_glyph(width = 0.028, height = 0.012) +
  scale_colour_manual(values = ...)
```



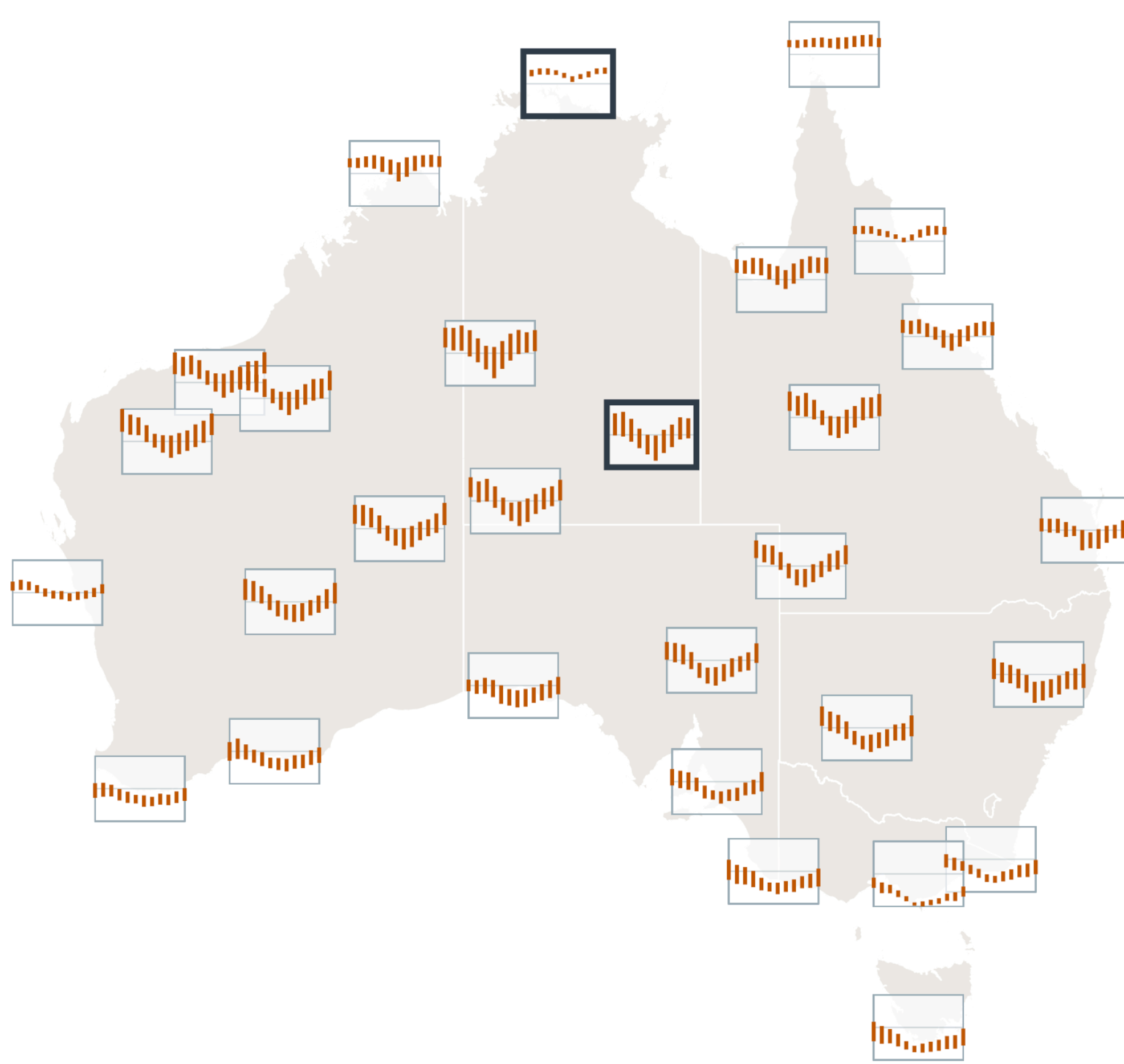
- Ocean beaches** spike the day after a storm (46% of samples unsafe), then clear within 2 days (24%).
- Harbour** sites are worse and slower: 73% unsafe the day after, still 65% two days later, especially in the inner-harbour coves.

2026 sugarglider: interval segments

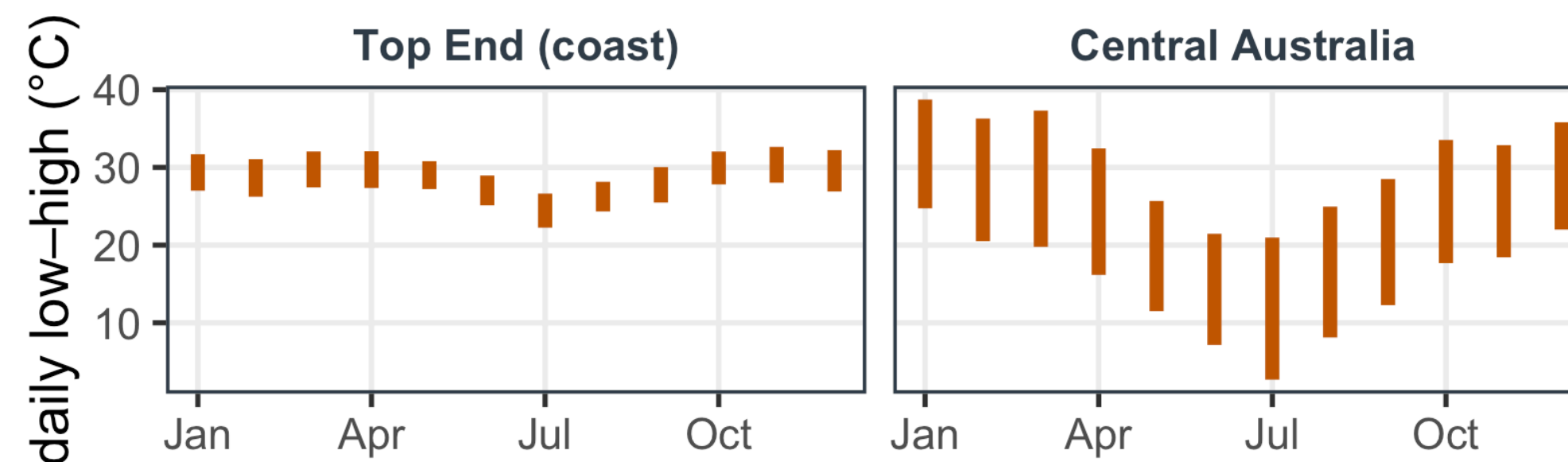
A line glyph shows **one value** per time point. Many summaries are an **interval**: a lower and an upper value of the same measurement, such as the daily low and high, or the quietest and busiest month. **sugarglider** adds aesthetics for the two ends of the interval, which share one scale inside each glyph.

Segments draw one bar per time point. Australian weather stations, monthly average daily low (bottom) to high (top), 2020 (`sugarglider::aus_temp`):

```
ggplot(aus_temp, aes(x_major = long, y_major = lat,
  x_minor = month,
  y_minor = tmin, yend_minor = tmax)) +
  add_glyph_boxes() + geom_glyph_segment()
```



The two boxed glyphs enlarged:



The **Top End** stays between 22 and 33 °C all year, with only 4–5 °C between day and night. **Central Australia** swings from 3 °C winter nights to 39 °C summer days, with 14–18 °C between day and night.

Take-home

- Glyph maps show space and time together.** Each location's time series is drawn where it belongs, so local patterns and their spatial structure can be explored in a single plot during exploratory analysis.
- Glyph maps are now part of ggplot2.** cubble provides line glyphs and sugarglider adds interval glyphs (segments and ribbons), and both combine with base maps, scales, colour and facets like any other layer.
- Glyph maps can be interactive.** The same glyphs can be placed on a leaflet map, so viewers can zoom into dense areas and look up individual locations.

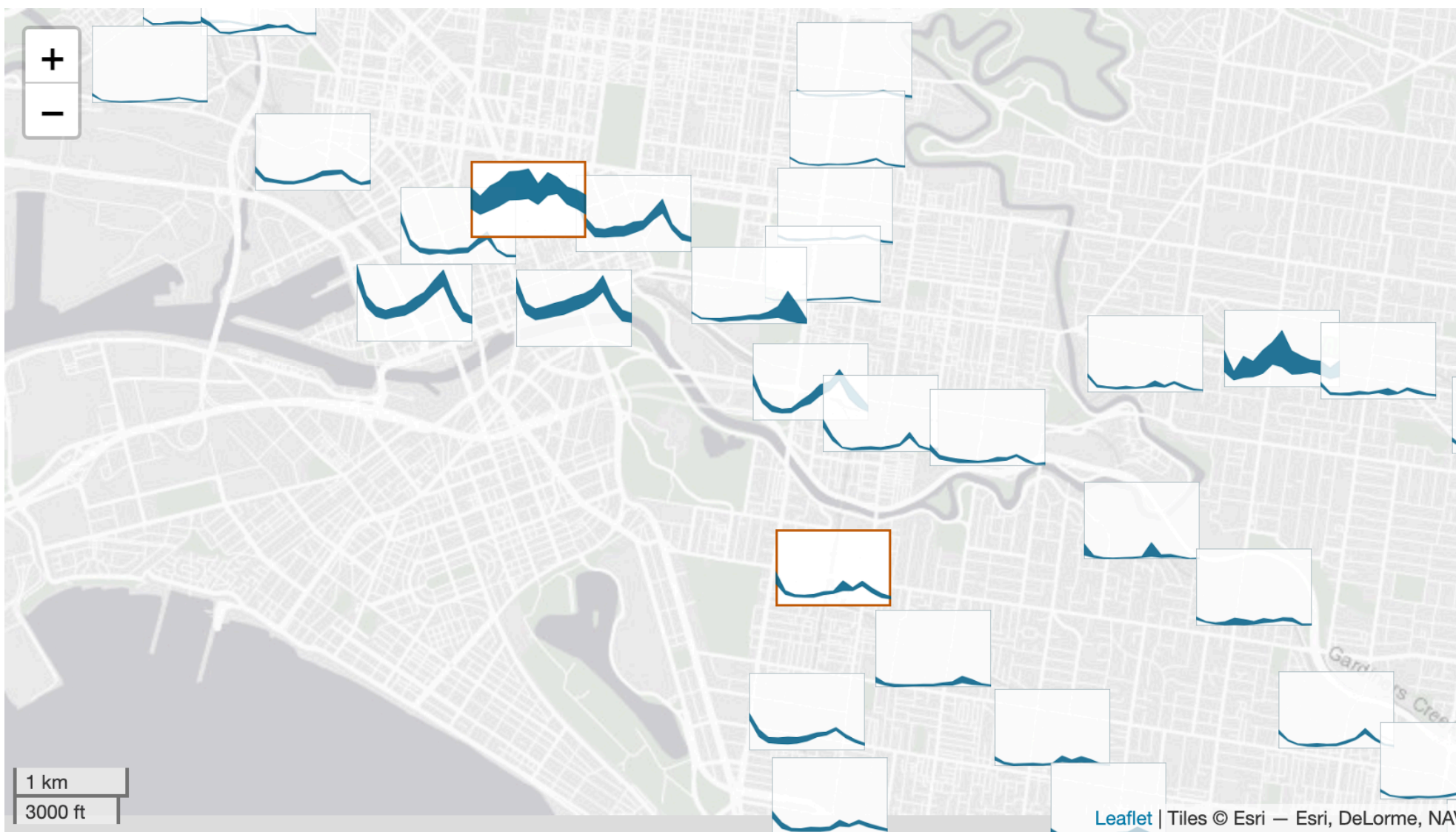
2026 sugarglider: interval ribbons

Ribbons fill the band between the lower and upper values. Melbourne train stations, hourly weekday passengers, 2023–24 (`sugarglider::train`), on an interactive **leaflet** map. For each hour, the **lower edge** is the passenger count in the quietest month and the **upper edge** in the busiest month. Each station's glyph is saved as an image and added as a leaflet marker (simplified):

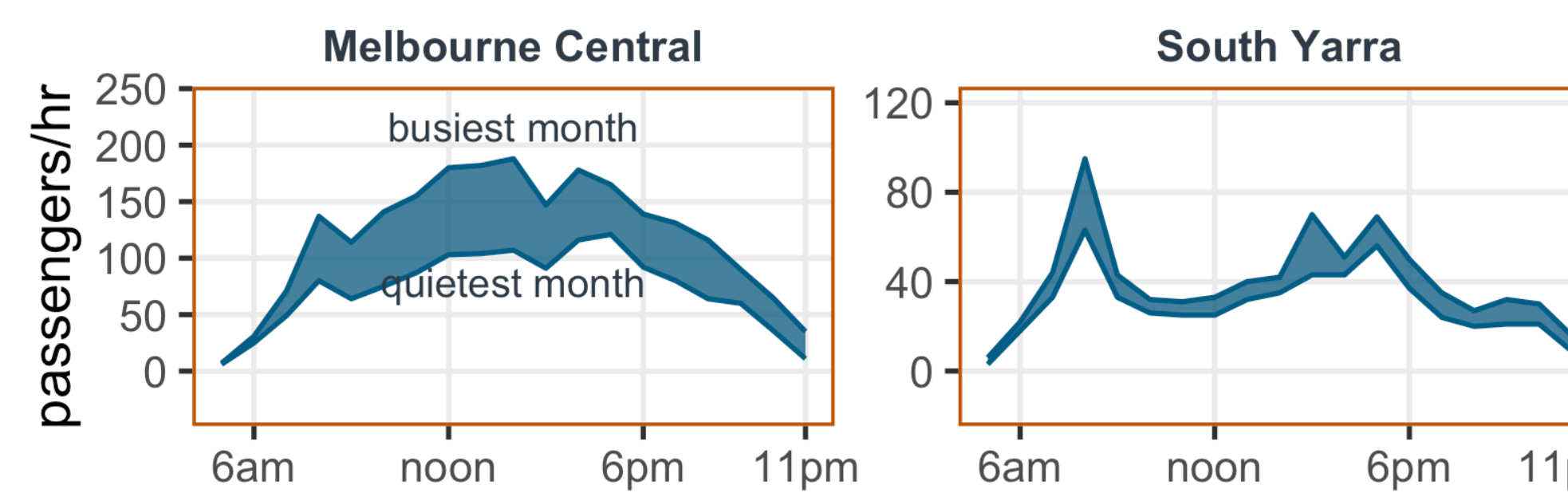
```
glyph <- ggplot(station,
  aes(x_major = long, y_major = lat, x_minor = hour,
      ymin_minor = min_weekday, ymax_minor = max_weekday)) +
  add_glyph_boxes() + geom_glyph_ribbon() + theme_void()
ggsave("glyph.png", glyph, bg = "transparent")

leaflet() |> addProviderTiles("Esri.WorldGrayCanvas") |>
  addMarkers(lng, lat, label = name,
    icon = makeIcon("glyph.png", 66, 44))
```

All stations share one scale, so glyph height shows how busy a station is; zoom in to separate crowded glyphs and hover for station names:



The two boxed glyphs enlarged, with their own axes:



City Loop stations carry by far the most passengers. On their own axes, suburban stations such as **South Yarra** show two sharp **commuter peaks**, while **Melbourne Central** is as busy at midday as at the peaks.

Questions? Email hsherryzhang@utexas.edu



Wickham, Hofmann, Wickham & Cook (2012). Glyph-maps for visually exploring temporal patterns in climate data and models. *Environmetrics* 23(5), 382–393. [doi:10.1002/env.2152](https://doi.org/10.1002/env.2152)
Zhang, Cook, Laa, Langrené & Menéndez (2024). cubble: An R package for organizing and wrangling multivariate spatio-temporal data. *J. Stat. Softw.* 110(7). [doi:10.18637/jss.v110.i07](https://doi.org/10.18637/jss.v110.i07)
Po, Yang, Zhang & Cook (2026). sugarglider: Create glyph-maps of spatiotemporal data. *The R Journal* 18(3), 286–304. [doi:10.32614/RJ-2026-050](https://doi.org/10.32614/RJ-2026-050)
Leaflet tiles: Esri. Beach data: [TidyTuesday 2025-05-20](https://tidytuesday.github.io/2025-05-20/).